How-To Library

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Getting Started with Data Domains

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Abstract

You can assign a data domain to a column or field asset to easily identify and understand the asset in Enterprise Data Catalog. This article describes how to create a data domain, enable data domain discovery for a resource, and view inferred data domains in Enterprise Data Catalog.

Supported Versions

- Enterprise Data Catalog 10.4.1

Table of Contents

Overview. 2
Background. 2
Lesson 1: Create Data Domains. 3
Lesson 2: Enable Data Discovery for a Resource 7
Lesson 3: Accept the Inferred Data Domains and View the Assigned Data Domains 8

Overview

Use data domains to classify data. A data domain contains a set of user defined rules to classify data objects like columns or tables based on metadata or data. You can also use data domains to identify and understand the meaning of critical source data or undiscovered source data to work effectively on it.

Background

HypoStores, a multinational organization in New York, recently acquired a new organization based out of Chicago. You are one of the data analysts in HypoStores. You need to analyze the spending patterns of the high-profile customers in both the organizations using the credit card information within different customer tiers.

The customer details are present across multiple tables and different databases such as Oracle and MySQL. You need to share your findings with the Chief Marketing Officer. He would use these results to send out exciting offers to the eligible customers of HypoStores.
The following table lists some of the challenges you might come across, along with possible solutions that you can achieve using Enterprise Data Catalog:

Challenge	Solution
You need to browse multiple sources in the two organizations to identify relevant data.	Identify the data sources that need to be scanned to find the required customers that match the eligibility criteria. The data sources are spread across multiple databases such as Oracle and MySQL. Create resources in Catalog Administrator to extract metadata from these data sources.
You cannot find and analyze the relevant columns that contain credit card information because of inconsistency in the column names and column data.	Enable domain discovery for the resources created in the data catalog. Create data domains to search for the required information across the data sources.

In this tutorial, you complete the following lessons on creating and using data domains:

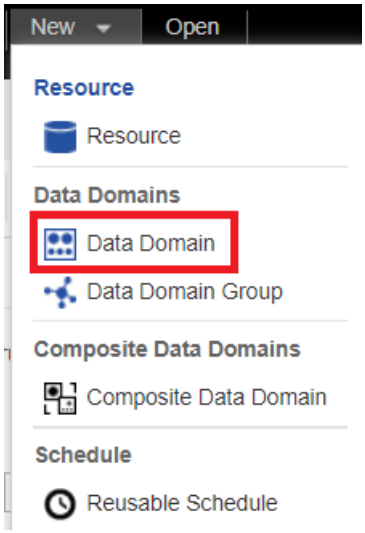
Lesson	Description
Create Data Domains	Create the following data domains to search for the required information in an Oracle resource named **Customer_Details** in Catalog Administrator. - CC_Number - CustomerID - CustomerTier
Enable Data Discovery for a Resource	Enable data discovery for the **Customer_Details** resource in Catalog Administrator.
Accept the Inferred Data Domains and View the Assigned Data Domains	Accept **CustomerID** and view **CC_Number** data domains for **Customer_Details** resource in Enterprise Data Catalog.

Lesson 1: Create Data Domains

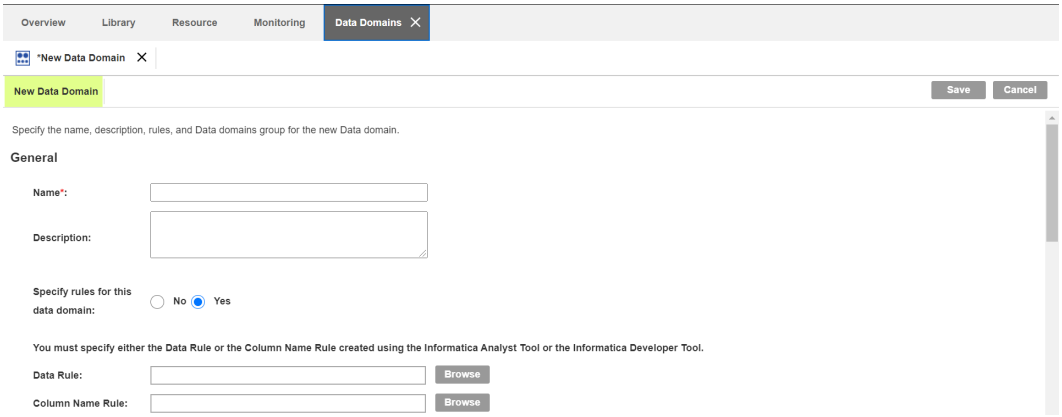
In this lesson, you will learn how to create a data domain so that you can find the relevant credit card information spread across multiple databases.

Perform the following actions to create data domains:

1. Log in to the Catalog Administrator.
2. Click **New** > **Data Domain**.



The **New Data Domain** page appears as shown in the following image:



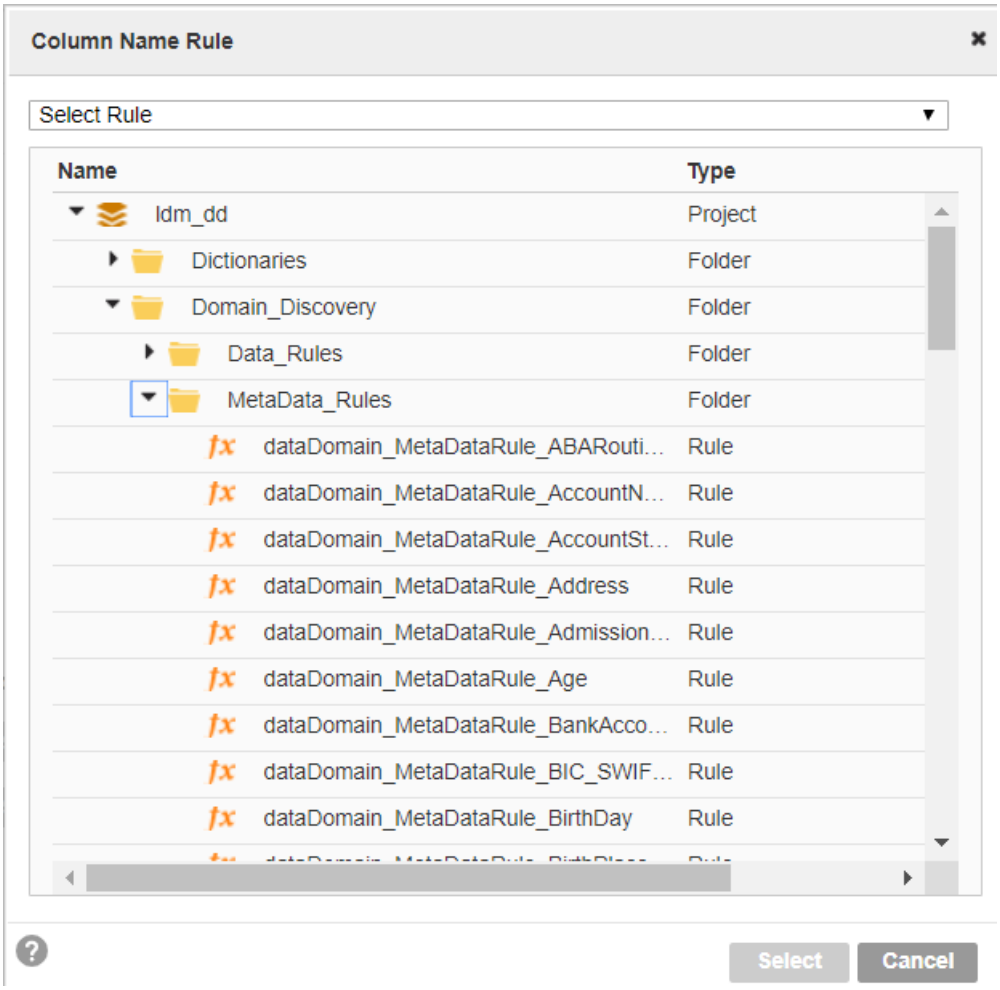
3. Enter the following details:

   - Name. CC_Number.

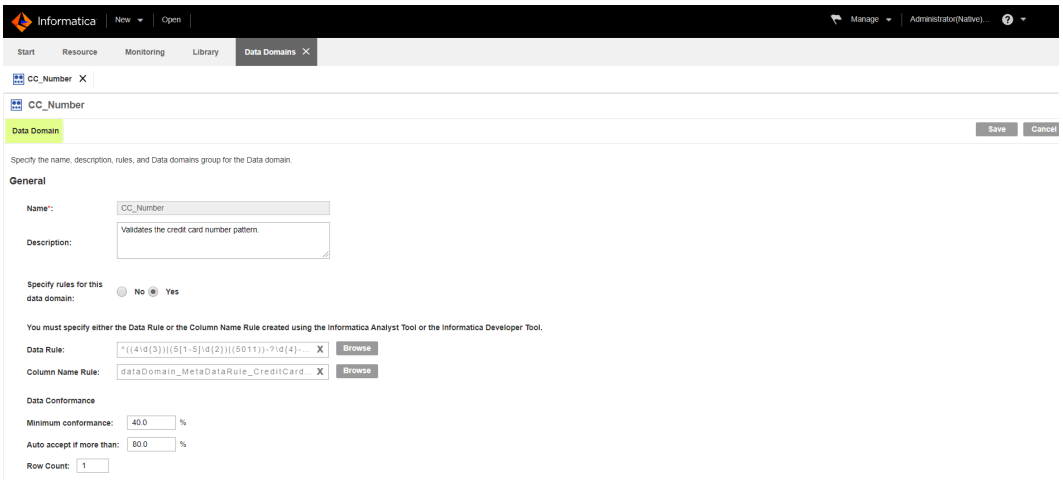
     Note: Enterprise Data Catalog comes with a set of predefined data domains including the **CreditCardNumber**. In this lesson, we create a data domain named **CC_Number** instead of using the predefined data domain.

   - Description. Validates the credit card number pattern.
   - By default, **Specify rules for this data domain** option is set to **Yes**.
   - Data Rule. Click **Browse** to choose one of the rules. To identify the credit card pattern, specify the following expression: `^((4\d{3})|(5[1-5]\d{2})|(5011))-?\d{4}-?\d{4}-?\d{4}|3[4,7]\d{13}$`
   - Column Name Rule. Click **Browse** to choose one of the rules. Choose **Select Rule** > **Idm_dd** > **Domain Discovery** > **MetaData_Rules** > **dataDomain_MetaDataRule_CreditCardNumber**.

     Note: The CreditCardNumber selected in the column name rule is one of the rules that is shipped by default. You can create your own rules based on regex, reference table, or mapplets.

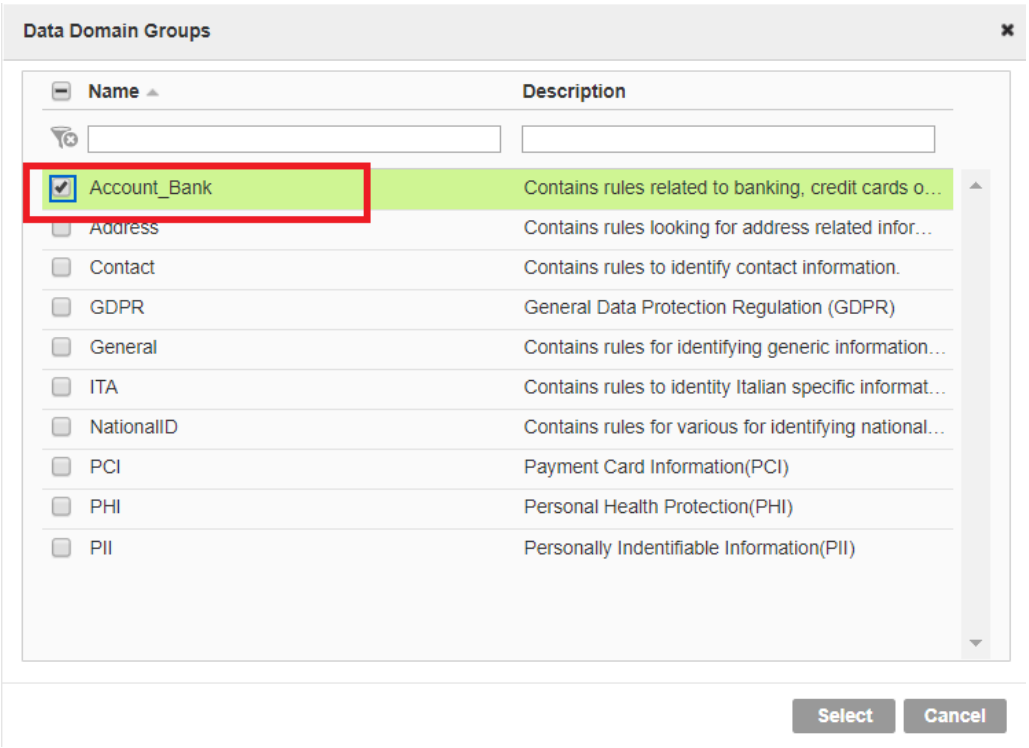


- Data Conformance. Enterprise Data Catalog identifies the data domains based on the conformance criteria. In this lesson, we will go with the following default values.
  - Minimum Conformance. The minimum conformance percentage for the data domain to be listed in the Enterprise Data Catalog. The default value is 40%.
  - Auto accept if more than. Enterprise Data Catalog accepts the data domain automatically if the data domain match exceeds the configured percentage. The default value is 80%.
  - Row Count. Displays the minimum number of rows for data conformance. The default value is 1.



4. Select the data domain groups to which you want to add the data domain.

   - Click  in the **Data Domain Groups** pane.
     The **Data Domain Groups** window appears.
   - Select the data domain group **Account_Bank** and click **Select**.



5. Click **Save**.

6. Similarly, create the following data domains:

Data Domain Name	Description
CustomerID	Customer Identifier. Enter `dataDomain_MetaDataRule_UniqueIdentifyingNumber` in the **Column Name Rule** field.
CustomerTier	Tiered programs are customer loyalty programs that give different rewards based on what tier a customer is in. The tiers are measured by dollars spent. The customer tiers are GOLD, SILVER and BRONZE. Enter `(?i)(?:[\s]|^)(Gold|Diamond|Ruby|Emerald|Silver|Bronze)(?=[\s]|$)` in the **Data Rule** field.

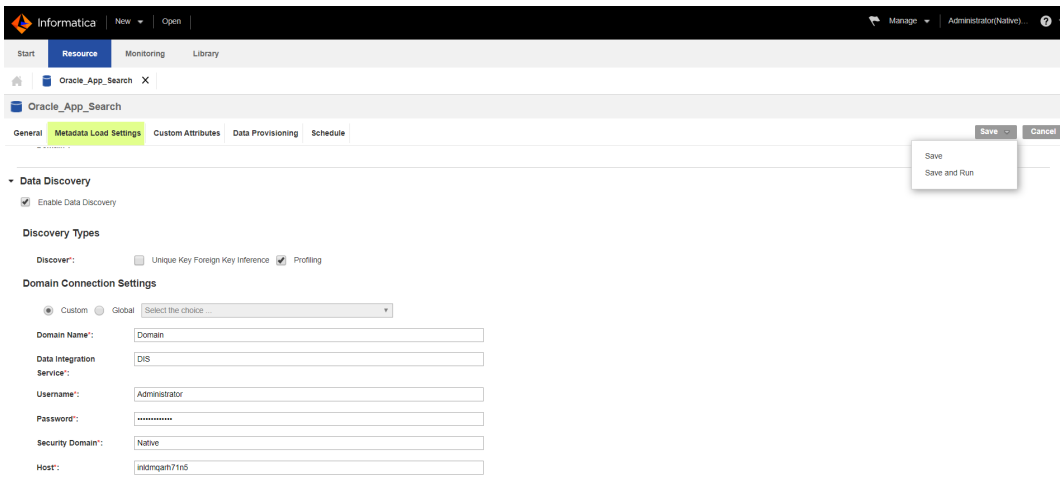
After creating the three data domains, the details are saved in the Model repository.

Lesson 2: Enable Data Discovery for a Resource

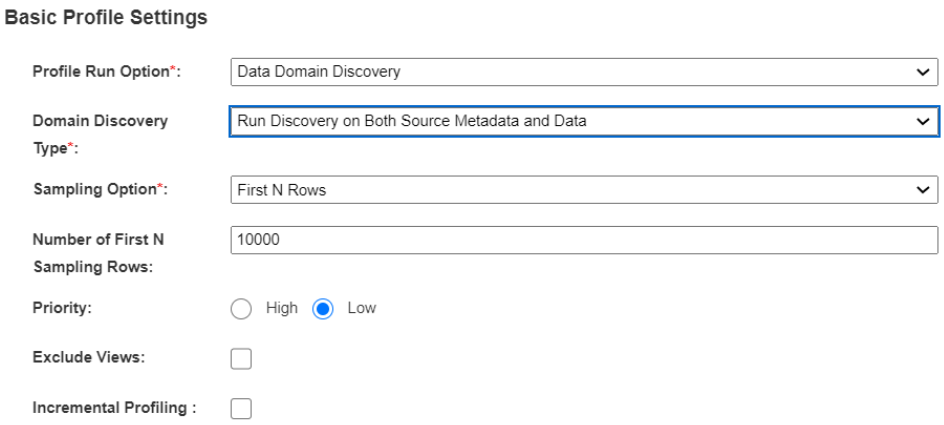
You can select a resource in the Catalog Administrator for Oracle or MySQL databases. You might also create a resource.

In this lesson, you will learn how to enable data discovery for the **Customer_Details** resource.

1. On the **Metadata Load Settings** page, select the **Enable Data Discovery** option and select the required data domain.



2. Alternatively, select the data domain group as required under the **Basic profile Settings** for the resource.



3. Click **Save and Run**.

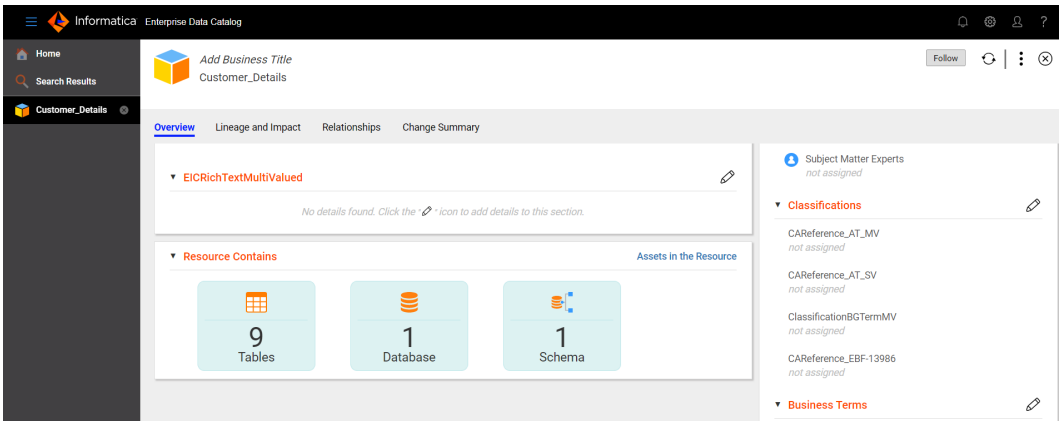
Lesson 3: Accept the Inferred Data Domains and View the Assigned Data Domains

Enterprise Data Catalog displays all the inferred and assigned data domains in the Data Domains list.

In this lesson, you will learn how to accept inferred data domains and view assigned data domains for the resource you had configured the data domains.

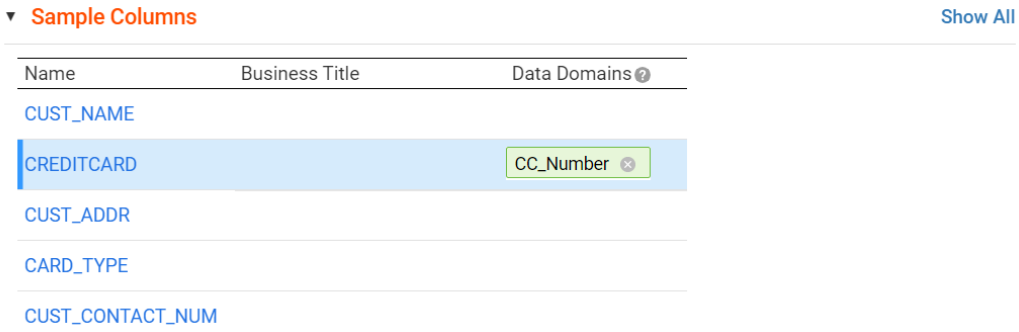
1. Log in to the Enterprise Data Catalog and search for the resource named **Customer_Details**.

   The tables, database, and schemas associated with the resource are listed.

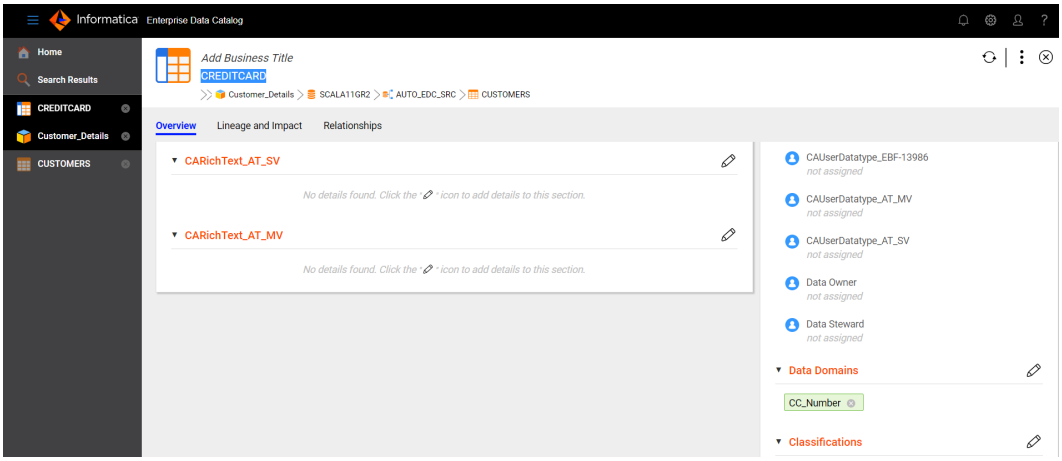


2. Click **Tables** from the **Resource Contains** section.
3. Select the table named **Customers** from the search results page.

The **Sample Columns** section displays all the columns tagged to this table.



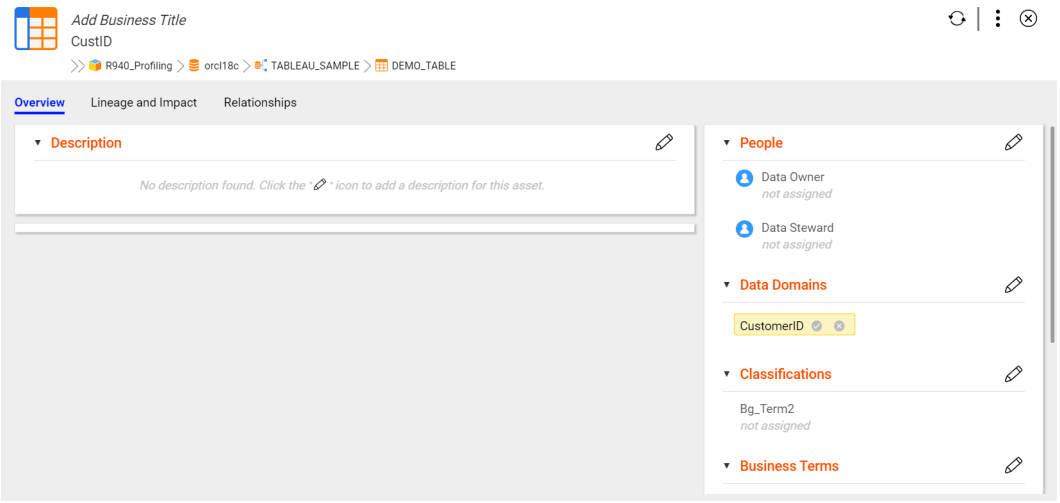
4. Select the column named **CREDITCARD** and on the right-hand side, the inferred data domain appears as shown in the following image:



Enterprise Data Catalog uses the data conformance properties that you configured while creating the data domains and lists the tables and columns that matches the conformance criteria.

Based on the conformance criteria, the columns are listed, and you can accept or reject the data domain accordingly. Here, the data domain is accepted as the data conformance was more than 80%.

5. Select the column named **Cust_ID** and on the right-hand side, the recommended data domain appears as shown in the following image:



   Here, based on the conformance criteria defined in the data domains, the columns are listed, and you can accept or reject the data domain **CustomerID** accordingly.

You are all set to provide the required information to the Chief Marketing Officer.

Author

Sabha S